DBA-RMCL: Refined Metric Contrastive Learning With Dual-Branch Attention for Few-Shot Remote Sensing Image Scene Classification

Ruchan Dong, Jinwei Xia, Licheng Jiao, *Fellow, IEEE*, Jin Zhao, Lei Hong, and Zengxin Yun

***Abstract*— Remote sensing scene classification plays a vital role in identifying Earth's surface features. However, traditional methods necessitate vast quantities of labeled training data, which significantly limits their applicability when labeled samples are scarce. To overcome this limitation, remote sensing few-shot classification (RSFSC) methods have arisen as a promising approach to maintaining classification accuracy with minimal data. In this article, we present a refined metric contrastive learning framework with dual-branch attention (DBA-RMCL), a novel framework that tackles two fundamental challenges in RSFSC: intraclass diversity and interclass similarity. Our framework employs a dual-branch architecture consisting of way and shot branches to reduce intraclass variance while enhancing feature consistency and interclass separability. The architecture incorporates feature fusion between branches to optimize feature learning and classification performance. We further enhance feature representations through our refined Siamese contrastive learning representation (RSCLR) module, which implements self-supervised contrastive learning. The framework also includes a combined attention-based prototypical head (CAP-Head) module with a compositional attention computation unit to model complex relationships between query sets and prototypes. Experimental results on six publicly available datasets demonstrate that DBA-RMCL achieves state-of-the-art performance compared to existing FSC methods. For 5-way 1-shot tasks, our framework attains accuracies of 77.15% ± 0.34% on NWPU-RESISC45 and 87.33% ± 0.22% on WHU-RS19. Additionally, we established the first benchmark for few-shot scene classification, MEET-FSL, using the MEET dataset. These outcomes not only underscore the efficacy of our dual-branch architecture but also represent a significant advancement in the field of few-shot learning for remote sensing classification.**

***Index Terms*— Contrastive learning, dual-branch framework, few-shot learning, image scene classification, remote sensing.**

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I. Introduction

REMOTE sensing imagery serves as a crucial source of geospatial data, providing comprehensive information about Earth's surface features and land-cover patterns. The automated classification of these images has become a cornerstone technique in Earth observation, enabling the systematic categorization of surface features from satellite and aerial imagery. This fundamental capability underpins numerous applications, including land-cover classification [1], change detection [2], object detection [3], image annotation [4], image retrieval [5], and other techniques [6], [7], [8], [9], [10], [11]. Yet, despite these advances, the inherent complexity and variability of Earth's surface features present significant challenges, particularly in acquiring sufficient labeled training data. Traditional classification approaches often falter when confronted with limited training samples, creating a pressing need for more innovative solutions that can perform effectively with smaller datasets [12], [13], [14].

Few-shot remote sensing image classification methods aim to accurately classify remote sensing images with limited labeled data [15], [16], [17], [18], [19], [20], [21], [22]. Compared with traditional methods, these techniques maintain high classification accuracy even with minimal annotated samples. Among the various approaches to few-shot learning, metric learning has garnered significant attention due to its low computational complexity, superior performance, and reduced dependency on data resources for new tasks. For instance, Gao et al. [23] proposed the LATA-BP-Bhattacharyya distance covariance (BDC) method, which effectively addresses challenges such as limited training samples, background interference, and the dual issues of interclass similarity and intraclass diversity by employing a Bi-Prototype BDC metric network. Qin et al. [24] introduced the FSRSSC method, which uses class subspaces as the metric basis to capture class commonalities, thereby mitigating the negative impact of irrelevant objects on classifier performance. Liu et al. [25] designed the CSPSR module, which leverages the rich annotated information in the query set to adaptively learn differentiated update rules for prototypes across different classes, further enhancing classification effectiveness.

In recent years, contrastive learning has been widely applied to few-shot learning tasks, achieving remarkable success [26], [27], [28], [29], [30]. The concept of contrastive learning

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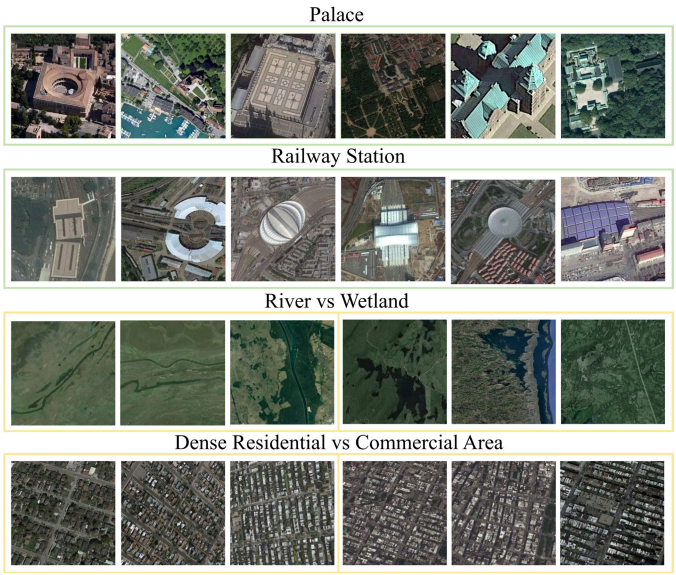


Fig. 1. Illustration of intraclass variance and interclass similarity challenges in remote sensing imagery from the NWPU-RESISC45 dataset. Green boxes demonstrate significant visual variations in the same object class (intraclass variance), while yellow boxes show cases where different object classes exhibit similar visual characteristics (interclass similarity). These classification challenges are particularly pronounced in few-shot learning scenarios with limited training samples.

was first introduced by Hadsell et al. [31], where the core idea is to maximize the similarity between samples of the same class while minimizing the similarity between samples of different classes. This enables the model to learn representations with higher intraclass similarity and excellent interclass separability. In self-supervised learning, contrastive learning often relies on proxy tasks [32], [33], [34] to drive the training process. Leveraging the superior performance of these methods, few-shot learning has also made significant advancements in remote sensing image classification. Ji et al. [28] designed two auxiliary supervised contrastive learning branches to enhance the model's ability to distinguish interclass differences while improving robustness to intraclass variations. Li et al. [35] proposed a self-supervised strategy based on contrastive learning to discover category-invariant features hidden in new class data. Xu et al. [30] developed a dictionary-based contrastive loss module (DBCL) that further improves interclass separability, providing a more effective solution for few-shot learning tasks.

Despite the significant progress achieved by contrastive learning-based few-shot remote sensing image classification methods, several challenges remain, particularly those related to intraclass diversity and interclass similarity, which are especially pronounced in remote sensing classification tasks. As illustrated in Fig. 1, remote sensing images of the same class may exhibit substantial intraclass variability due to scene complexity, camera angles, and the inherent diversity of object categories. Simultaneously, the high similarity between different classes further exacerbates classification difficulty. These issues are particularly critical in few-shot learning scenarios because the methods rely on extracting and distinguishing features with limited samples, making the requirements for feature extraction and discrimination more stringent. Consequently, the challenges of intraclass variability and interclass similarity become even more prominent, posing significant barriers to classification performance. Designing more effective metric strategies and contrastive learning methods under the few-shot learning framework to enhance feature discrimination remains a core research challenge. Addressing this issue is crucial to improving the robustness and accuracy of few-shot remote sensing image classification models.

Dual-branch architectures have demonstrated remarkable performance in deep learning across various tasks [36], [37], [38], particularly in handling multimodal data and contrastive learning tasks. In multimodal learning, dual-branch architectures are widely used to process information from different modalities (e.g., image and text). Each branch focuses on a specific modality, extracting its unique features, and the final feature fusion stage integrates information across modalities [39], [40], [41]. Similarly, in contrastive learning, dual-branch architectures operate on different data augmentations of the same input. Each branch processes a distinct augmented view, and the contrastive loss encourages the model to learn more discriminative features [42], [43], [44]. Inspired by the principles of dual-branch architectures, we propose an innovative approach where each branch addresses a specific challenge: one focuses on reducing intraclass variability, while the other aims to eliminate interclass similarity. By leveraging this dual-branch structure, the model effectively tackles both challenges by processing intraclass and interclass features separately. The advantages of each branch are subsequently combined during the feature fusion stage, leading to improved classification performance.

Drawing on advances in contrastive learning and dual-branch architectures, we present a refined metric contrastive learning framework with dual-branch attention (DBA-RMCL), a novel framework for few-shot remote sensing image classification. DBA-RMCL specifically addresses two key challenges in few-shot learning: intraclass feature diversity and interclass feature similarity. The framework consists of three core modules: refined Siamese contrastive learning representation (RSCLR), dual-branch adaptive combine attention (DBA-CA), and combined attention-based prototypical head (CAP-Head). The RSCLR module builds on SimSiam [45] to maximize intraview similarity while minimizing interview distribution differences across augmented views, resulting in more advanced feature representations. The DBA-CA module employs a scale-adaptive dual-branch structure that processes the support set to capture both intraclass and interclass relationships. The CAP-Head module incorporates stackable combine attention unit (CAU) that integrate self-attention and cross-attention layers, enabling sophisticated feature interactions between query sets and prototypes. We evaluated DBA-RMCL on six public datasets: NWPU-RESISC45 [46], WHU-RS19 [47], UC-Merced [48], AID [49], EuroSAT [50], and MEET [51]. Our experimental results demonstrate state-of-the-art performance, surpassing both classical few-shot learning methods and current metric-based approaches in remote sensing image classification. These results validate both the effectiveness of DBA-RMCL and the advantages of our dual-branch architecture in addressing feature-related challenges.

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The key contributions of this work include the following.

1) *A Novel Integrated Framework:* We introduce DBA-RMCL, which combines contrastive learning with dual-branch architecture to address few-shot remote sensing classification challenges. The framework's three modules—RSCLR, DBA-CA, and CAP-Head—work in concert to enhance classification performance.
2) *Advanced Feature Learning:* Our RSCLR module leverages contrastive learning across augmented image views, maximizing intraview similarity while minimizing cross-view distribution differences. This approach yields more advanced and diverse feature representations for downstream tasks.
3) *Dual-Branch Feature Processing:* The DBA-CA module's way and shot branches effectively distinguish between classes while maintaining intraclass consistency. The way branch enhances interclass discrimination, while the shot branch refines intraclass attention weights, improving overall feature discrimination.
4) *Enhanced Prototype–Query Interaction:* The CAP-Head module's innovative CAU optimizes prototype–query feature interactions, enabling fine-grained similarity detection and improving classification accuracy.

The remainder of this article is organized as follows. Section II reviews related work in the field of few-shot learning for remote sensing image scene classification. Section III details the proposed methodology. Section IV presents the experimental setup and results. Finally, Section V concludes the article and discusses potential future directions.

II. Related Work

This section briefly overviews image classification in the context of few-shot and contrastive learning. We also discuss relevant work on self-attention mechanisms in few-shot learning.

A. Few-Shot Learning for Remote Sensing Images

The acquisition and annotation of remote-sensing images are often costly, time-consuming, and challenging. As a result, few-shot learning has garnered widespread attention in remote sensing as a practical approach for enhancing classification performance using a limited number of labeled remote sensing data. Few-shot learning can be categorized into classic training and scene training. Classic training typically utilizes transfer learning, leveraging pretrained models on large-scale datasets to extract high-level features [27], [52]. However, due to data distribution discrepancies, models are prone to overfitting and catastrophic forgetting. Scene training divides the dataset into multiple tasks, each consisting of a support set and a query set, using a multitask learning strategy to enhance model generalization and alleviate catastrophic forgetting [53], [54].

Metric learning refers to training models to learn an effective and discriminative feature representation or similarity metric from limited samples. Cheng et al. [55] proposed D-CNN, which optimizes a discriminative objective function and a metric learning regularization term. This approach ensures that images from the same scene category are mapped closer together while pictures from different categories are pushed farther apart, enhancing the model's discriminative capability. Shi et al. [56] introduced a semi-supervised classification framework based on improved metric learning and convolutional neural networks, transforming the vector distance parameter problem into a neural network design problem. This framework alleviates difficulties in similarity prediction between high-dimensional features by employing backpropagation (BP) algorithms to compute parameters automatically. Li et al. [57] proposed RS-MetaNet, which uses metric spaces for remote sensing scene classification. This method maximizes the distance between different categories, enhancing the model's generalization ability to new samples while ensuring proper fitting and providing better linear decision boundaries for various categories. Additionally, Jiang et al. [58] introduced multiscale metric learning (MSML) for extracting multiscale features and learning multiscale relationships between samples. Higher-level features correspond to deeper metric learning, while lower-level features correspond to lighter metric learning. Metric learning generally represents the relationship between different image categories through similarity measures without needing large labeled datasets or pretrained models, thus improving the model's robustness and generalization ability.

However, these few-shot learning methods often face challenges in effectively leveraging the complex semantic relationships within and between classes. This can lead to reduced efficiency and accuracy in remote sensing scene classification. Our proposed approach addresses these limitations through the DBA-CA module, specifically designed to handle the intricate relationships between intraclass and interclass features. The dual-branch architecture enhances the compactness of intraclass features while increasing the separability of interclass features, thereby enabling the extraction of more discriminative feature representations for adequate classification.

B. Contrastive Learning

Traditional classification algorithms often suffer from overfitting. In contrast, contrastive learning mitigates this issue by comparing features of similar and dissimilar samples, enabling the model to focus on interclass differences, thereby reducing overfitting risk and enhancing generalization performance. Bi et al. [26] proposed ViT-CL, which extends self-supervised contrastive learning to fully supervised scenarios, improving model robustness and providing better control over the optimization process through a two-stage optimization framework. Li et al. [27] introduced MMPC-Net, which integrates multiscale and multigranularity representation techniques to extract more discriminative features. By employing a progressive contrastive learning module, MMPC-Net effectively captures subtle differences across features of varying scales, successfully identifying nuanced variations within positive and negative sample pairs. Ji et al. [28] proposed the DCN, incorporating two auxiliary contrastive learning branches that independently focus on interclass separability and intraclass invariance, enabling the model to learn richer discriminative features.

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Despite the advancements in contrastive learning, its application to remote sensing image classification still faces several challenges. Traditional contrastive learning methods often rely on complex network architectures or require additional training steps to construct positive and negative sample pairs, which significantly increases the complexity of model training. Yang et al. [29] proposed ICSFF, which performs further optimization through a self-supervised approach following supervised pretraining. However, this method necessitates additional pretraining steps, increasing training time and computational overhead.

In contrast to these methods, we propose the DBA-RMCL framework, which directly performs self-supervised contrastive learning through a twin residual network, eliminating the need for additional pretraining steps. The RSCLR module within DBA-RMCL achieves high-quality feature representation by maximizing the similarity between different augmented views of the same image while minimizing the distributional differences of the same augmented view across various batches. This approach simplifies the training process, avoids the construction of positive and negative sample pairs, and reduces additional training overhead.

C. Self-Attention Mechanism

The self-attention mechanism has widespread applications in deep learning, demonstrating remarkable feature extraction capabilities, particularly in natural language processing and computer vision. Its core concept involves capturing global dependencies by computing the relevance between any two positions in a sequence, thereby enhancing the representational power of models. In few-shot learning tasks, the self-attention mechanism mitigates feature extraction deficiencies caused by insufficient samples. For instance, Li et al. [59] proposed SaberNet, which leverages self-attention to generate and refine features, inferring relationships between local features by comparing different patches to enhance detailed feature extraction. Similarly, Sun et al. [60] introduced SLTRN, which utilizes sample-level self-attention to strengthen the relational module's comparison capability by uncovering latent relationships among support classes. Moreover, Huang and Choi [61] developed SAPENet, which employs multihead self-attention to enhance the distinctiveness of sample features and generates intraclass channel attention maps to preserve key information. By integrating feature maps with attention maps, SAPENet constructs representative class prototypes.

While self-attention mechanisms demonstrate significant advantages in enhancing feature diversity and global consistency, they still face limitations in scenarios with limited samples. To achieve more accurate classification under such constraints, this study proposes a composite attention computation unit integrated into the CAP-Head module of the DBA-RMCL framework. This composite unit effectively captures complex feature interactions between query sets and prototypes, enabling the extraction of fine-grained similarity features and significantly enhancing classification performance.

III. Methodology

The DBA-RMCL framework addresses two critical challenges in remote sensing image classification: substantial intraclass variation and interclass similarity. The framework leverages the RSCLR module for self-supervised contrastive learning to extract high-level features from images. The DBA-CA module employs a dual-branch feature learning strategy to effectively manage the complex relationships between intraclass and interclass features, enabling the acquisition of accurate classification feature representations. Additionally, the composite attention computation unit within the CAP-Head module optimizes the feature representations of prototypes and query sets, ensuring that the prototype representations are fine-grained and distinctive.

A. Overview

To learn more advanced feature representations and enhance the consistency of intraclass features and the distinguishability of interclass features in few-shot remote sensing image classification tasks, we propose the DBA-RMCL, as shown in Fig. 2. Initially, we perform data augmentation on images using the SimSiam method [45], followed by feature extraction using a four-layer residual network, ResNet12 [62], to obtain feature representations of the images. Next, a multilayer perceptron (MLP) is employed as a prediction layer, where self-supervised contrastive learning is conducted by calculating the cosine similarity of the same image under different views, further extracting high-level feature representations. Subsequently, the support set features are input into the DBA-CA module for representation enhancement. Through its dual-branch architecture, the DBA-CA module optimizes the intraclass feature compactness and interclass feature separability, effectively improving the model's discriminative ability while maximizing the utilization of limited labeled data. Then, prototypes generated from the support set features are fused with query set features through the composite attention computation unit in the CAP-Head module, capturing the complex feature interaction relationships between the query set and the prototypes, thereby acquiring fine-grained similarities and enhancing the model's classification performance. Finally, classification is performed by computing the similarity between the query set and the prototypes.

B. Refined Siamese Contrastive Learning Representation

In this study, to fully leverage the limited image resources in few-shot learning and achieve the goal of obtaining effective, accurate, and high-level feature representations with a small amount of data, we propose a self-supervised contrastive learning representation method named RSCLR. The technique aims to maximize the similarity of the same image from different perspectives and minimize the distributional differences of the same augmented view across different batches, enabling the model to learn more advanced feature representations.

RSCLR consists of an encoder F and a nonlinear prediction layer. The encoder F is composed of ResNet12 and a nonlinear projection layer. During training, both the support set and query set are utilized simultaneously. For any image $x \in \mathbb{R}^{C\times H\times W}$ from the training dataset D_{train}, where C denotes the number of channels, H the image height, and W the image width, two random augmentations are applied to obtain x_1 and x_2. These augmented images are then passed through

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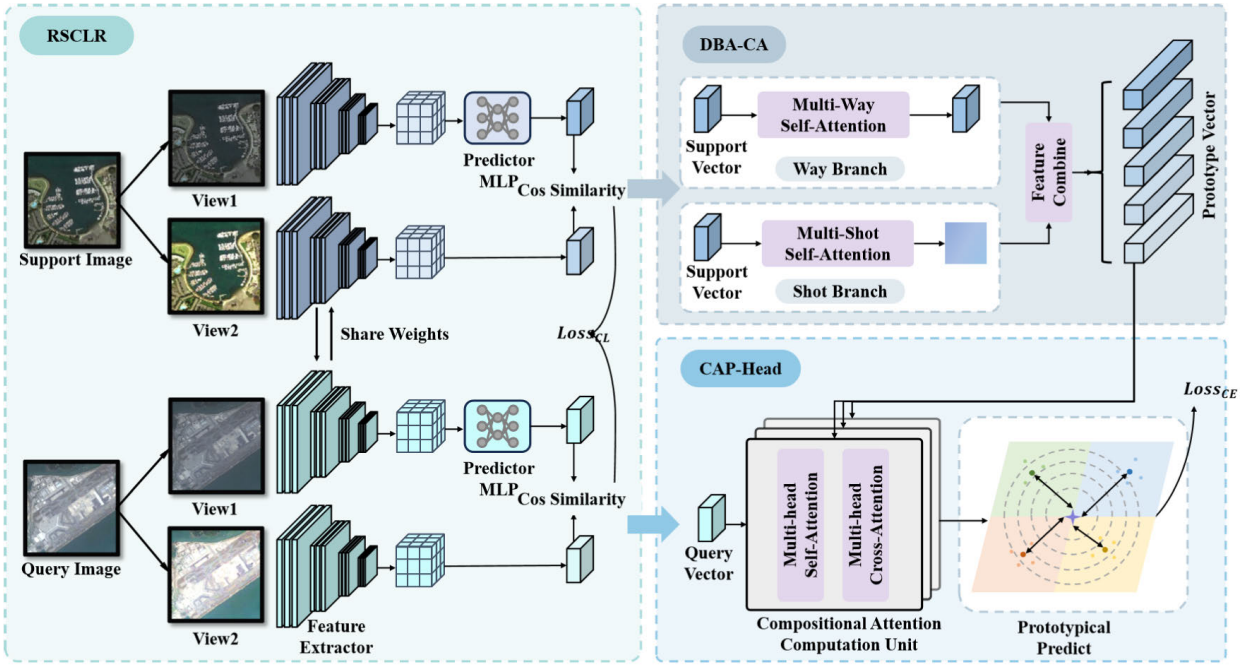


Fig. 2. Architecture of the DBA-RMCL framework for few-shot remote sensing classification. The framework consists of three main components: the RSCLR module implements self-supervised learning through image augmentation, extracting high-level features by computing cosine similarities between different views of the same image. The DBA-CA module processes support set feature vectors through two parallel branches. The CAP-Head module processes query set features through stacked attention units, fusing them with prototype features to model complex feature relationships. The framework is trained end-to-end by jointly optimizing self-supervised and supervised learning objectives, with the final classification determined by prototype–query similarity measurements.

the encoder to extract features $z_1, z_2 \in \mathbb{R}^{1\times D}$, where D represents the feature dimension. A nonlinear prediction layer is applied to obtain predictions p_1, $p_2 \in \mathbb{R}^{1\times D}$. To measure the similarity between the two vectors, we use cosine similarity, and the method is optimized to maximize the similarity of the same image under different views, as shown in the following equation:

$$E_{(x\sim X,\tilde{x}_1,\tilde{x}_2\sim T(x))}[\mathrm{sim}(z_1, z_2)]. \tag{1}$$

Here, X represents the image distribution, T represents the image augmentation distribution, and $\mathrm{sim}(\cdot,\cdot)$ denotes the cosine similarity between two feature vectors. To prevent the model from suffering from the "collapse solution" phenomenon during the unsupervised contrastive learning process, we introduce a symmetry loss function L_{cl} in the following equation:

$$L_{\mathrm{cl}} = -\frac{1}{2}\left[l(p_1, \mathrm{stopgrad}(z_2)) + l(p_2, \mathrm{stopgrad}(z_1))\right]. \tag{2}$$

The function $l(\cdot,\cdot)$ computes the loss, where the cosine similarity loss between two vectors is measured in the following equation:

$$l(p, \mathrm{stopgrad}(z)) = \frac{p}{\|p\|_2} \cdot \frac{\mathrm{stopgrad}(z)}{\|\mathrm{stopgrad}(z)\|_2}. \tag{3}$$

The stop grad operation truncates the z gradients to prevent the symmetry in the loss function from causing the model to converge to a collapsed solution. Through this design, RSCLR effectively maximizes the similarity between all augmented views of the same image while minimizing the distributional discrepancy of the same augmented view across different batches, thus obtaining high-quality feature representations. This method efficiently uses limited data resources and facilitates accurate feature representation in few-shot learning tasks. The expected value optimization for RSCLR is shown in the following equation, and the pseudocode for the implementation in PyTorch is provided in Algorithm 1:

$$E_{(x\sim X,\tilde{x}_1,\tilde{x}_2\sim T(x))}\left[l(p_1, \mathrm{stopgrad}(z_2)) + l(p_2, \mathrm{stopgrad}(z_1))\right]. \tag{4}$$

C. Dual-Branch Adaptive Combine Attention

In this study, we introduce the innovative DBA-CA module, which adopts a dual-branch structure designed to adaptively process support set features at different data scales for efficient capture of both interclass and intraclass relationships. A schematic of the module is shown in Fig. 3. Specifically, the way branch focuses on interclass differences, aiming to learn and extract discriminative features. This branch calculates the relationships between different classes by modeling the support set features, enhancing the model's ability to distinguish between classes. This process provides more apparent feature representations for subsequent classification decisions. On the other hand, the shot branch emphasizes intraclass diversity, exploring subtle differences among samples within the same class. By generating intraclass consistency attention weights, this branch captures similarities between intraclass samples and output feature weights by focusing on these similarities. This process effectively enhances the consistency of the support set features within each class, thereby improving the samples' expression and discriminative power. Finally, the features extracted by both branches are fused using element-wise multiplication, generating more decadent and more distinctive intraclass and interclass feature representations. This feature

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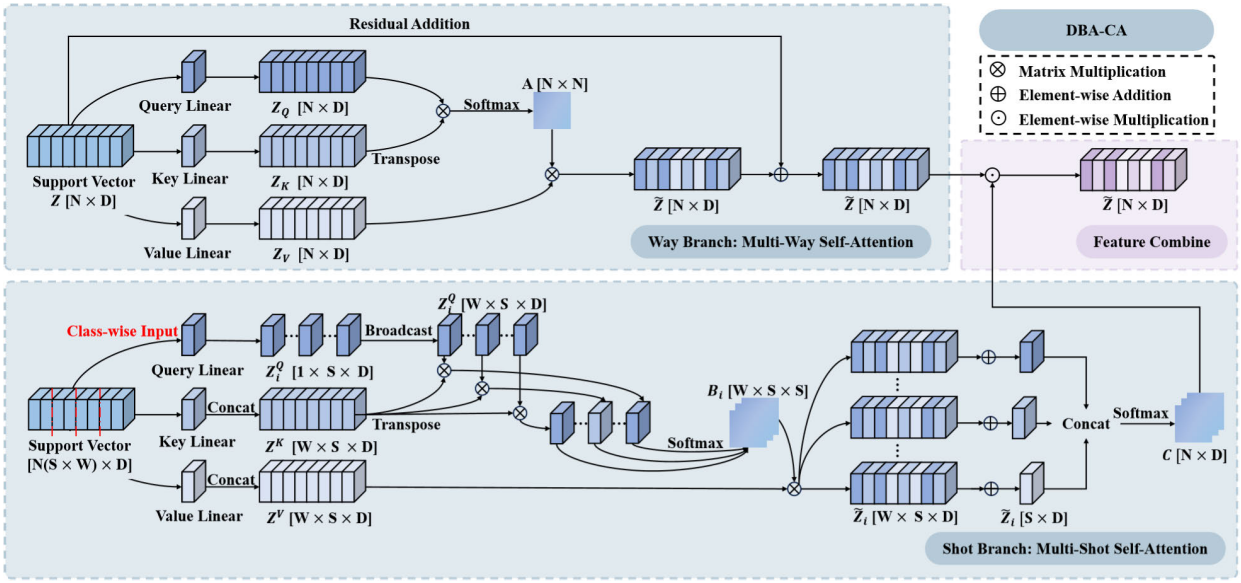


Fig. 3. Architecture of the DBA-CA module. The module consists of two branches: the way branch and the shot branch. The way branch primarily focuses on interclass differences, aiming to learn and extract discriminative features. The shot branch emphasizes intraclass diversity, exploring subtle differences within samples of the same class and generating an intraclass feature consistency attention map. Finally, the feature combine step fuses the features by element-wise multiplication, resulting in a more discriminative high-level feature representation.

Algorithm 1 Pseudocode of RSCLR, PyTorch-Like

```
# s: data in the support set
# q: data in the query set
# f: resnet12 + projection mlp
# h: prediction mlp

for s, q in loader do:
    x ← cat(s, q) # use both data in support and query set
    x_1, x_2 ← aug(x), aug(x) # random augmentation
    z_1, z_2 ← f(x_1), f(x_2) # projection head (n, d)
    q_1, q_2 ← h(z_1), h(z_2) # prediction head (n, d)

    L ← D(p_1, z_2)/2 + D(p_2, z_1)/2 # contrastive loss

    L.backward()
    update(f, h)
end for

function D(p, z): # negative cosine similarity
    z ← z.detach() # stop gradient

    p ← normalize(p, dim = 1) # l2-normalize
    z ← normalize(z, dim = 1) # l2-normalize
    return −(p · z).sum(dim = 1).mean()
end function
```

fusion strategy strengthens the model's ability to recognize class differences and improves its representation of intraclass diversity, further boosting classification performance.

1) Way Branch: We design a Multiway Self-Attention mechanism to capture interclass differences from a global perspective. For simplicity, we illustrate our self-attention block with a single attention head. Given a batch of support set images, let $Z = F(x)$ denote the feature map of these images, where $Z \in \mathbb{R}^{N\times D}$, with N representing the number of support set images and D the feature dimension. This feature map is projected into three distinct representations using three learnable linear projection layers: Q, K, and V. These projections generate new feature maps Z_Q, Z_K, and Z_V via the following linear transformations: $Z_Q = W_Q Z, \quad Z_K = W_K Z, \quad Z_V = W_V Z$, where W_Q, W_K, and W_V are learnable projection matrices. Next, we compute the self-attention score by performing a matrix multiplication between Z_Q and the transpose of Z_K, establishing the global relationship between features. This is mathematically expressed in the following equation:

$$\text{Attention Score} = Z_Q Z_K^{\mathrm{T}}, \quad \text{Score} \in \mathbb{R}^{N\times N}. \tag{5}$$

The self-attention scores are then normalized using the Softmax function along the sample dimension, which normalizes each attention score to produce the self-attention map $A = \text{stack}[z_0, z_1, z_2, \ldots, z_{N-1}] \in \mathbb{R}^{N\times N}$, as shown in the following equation:

$$z_i = \frac{\exp\left(z_Q^i \left(z_K^i\right)^{\mathrm{T}}\right)}{\sum_{i=1}^{N} \exp\left(z_Q^i \left(z_K^i\right)^{\mathrm{T}}\right)}, \quad z_i \in \mathbb{R}^{1\times N} \tag{6}$$

where N is the number of samples and i denotes the ith sample. Finally, the self-attention map A is multiplied by Z_V, and a residual connection is applied to ensure the features' stability and the information's effective propagation. This is mathematically expressed, as shown in the following equation:

$$\tilde{Z} = AZ_V + Z. \tag{7}$$

H is the scaling factor used in multihead attention, ensuring the balanced learning of features across different heads.

2) Shot Branch: We propose the Multishot Self-Attention mechanism to enhance the consistency and diversity of intraclass feature representations, encouraging the model to focus on self-attention modeling of support set samples within the

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same class. The Multishot Self-Attention shares the linear projection layer parameters used in the Multiway Self-Attention, ensuring spatial consistency of the feature maps and allowing the features from the two branches to be projected into similar feature spaces as much as possible. For the jth sample $x_{(i,j)}$ in class i, its feature representation is given by $Z_{(i,j)} = F(x_{(i,j)})$, where $Z_{(i,j)} \in \mathbb{R}^{1\times D}$. We generate the feature maps $Z^Q_{(i,j)}$, $Z^K_{(i,j)}$, and $Z^V_{(i,j)}$ through linear projections: $Z^Q_{(i,j)} = W_Q Z_{(i,j)}$, $Z^K_{(i,j)} = W_K Z_{(i,j)}$, $Z^V_{(i,j)} = W_V Z_{(i,j)}$, where W_Q, W_K, and W_V are learnable projection matrices. Next, we concatenate $Z^K_{(i,j)}$ and $Z^V_{(i,j)}$ for the same class samples along the sample dimension, resulting in Z^K_i and Z^V_i, which are defined as Z^K_i, $Z^V_i \in \mathbb{R}^{1\times S\times D}$, as shown in the following equation:

$$\begin{aligned} Z^K_i &= \mathrm{cat}\left[Z^K_{(i,1)}, Z^K_{(i,2)}, \ldots, Z^K_{(i,S)}\right], \quad \dim = 0 \\ Z^V_i &= \mathrm{cat}\left[Z^V_{(i,1)}, Z^V_{(i,2)}, \ldots, Z^V_{(i,S)}\right], \quad \dim = 0. \end{aligned} \tag{8}$$

Let W represent the number of classes and S the number of samples within the same class. To compute the self-attention scores, we broadcast $Z^Q_{(i,j)} \in \mathbb{R}^{1\times D}$ to align its first dimension with Z^K_i and Z^V_i. We then perform a matrix multiplication between $Z^Q_{(i,j)}$ and the transpose of Z^K_i to compute the intraclass self-attention scores. The resulting attention scores are normalized using the Softmax function along the class dimension, producing a set of intraclass attention maps $B = \{B_0, B_1, B_2, \ldots, B_W\}$, as shown in the following equation:

$$B_i = \frac{\exp\left(Z^Q_i \left(Z^K_i\right)^{\mathrm{T}}\right)}{\sum_{i=1}^{W} \exp\left(Z^Q_i \left(Z^K_i\right)^{\mathrm{T}}\right)}, \quad B_i \in \mathbb{R}^{W\times S\times S}. \tag{9}$$

Subsequently, we multiply each intraclass attention map B_i with Z^v to obtain the intraclass attention feature representation $\tilde{Z}_i$, as expressed in the following equation:

$$\tilde{Z}_i = [B_0 Z^v, B_1 Z^v, \ldots, B_W Z^v], \quad \tilde{Z}_i \in \mathbb{R}^{S\times D}. \tag{10}$$

For each $\tilde{Z}_i$, we sum these features element-wise along the class dimension to obtain the final intraclass attention feature $\tilde{Z}_i$. Finally, we stack all $\tilde{Z}_i$ and perform a normalization operation along the sample dimension to obtain the intraclass feature consistency attention map C, as shown in the following equation:

$$C = \frac{\exp(\mathrm{stack}(\tilde{Z}_i))}{\sum_{i=1}^{N} \exp(\mathrm{stack}(\tilde{Z}_i))}. \tag{11}$$

By leveraging the feature representations extracted through the way branch and shot branch, we effectively enhance intraclass consistency and interclass distinguishability. The feature combine module ultimately integrates these two feature representations through element-wise multiplication, producing a more expressive global feature representation. This fusion strategy effectively addresses the challenges of weak interclass distinguishability and significant intraclass variability.

D. Combine Attention-Based Prototypical Head

We propose the CAP-Head module based on the prototypical network to improve prototype prediction accuracy in few-shot learning tasks. Unlike traditional prototype prediction methods, CAP-Head introduces a combined attention computation unit before prototype prediction, further optimizing the prototype features' representation and prediction process. A schematic of the module is shown in Fig. 4. The CAP-Head module consists of L layers of CAUs, each comprising self-attention and cross-attention layers. In the CAP-Head module, query set features are first processed by the self-attention layer for feature relationship modeling. The output features are then used as queries, while the prototype features serve as keys and values, which undergo further feature fusion through the cross-attention layer. This design enables the CAP-Head module to effectively capture more complex semantic relationships between the query set and prototype features through the stepwise operations of the L-layer combined attention, enhancing the dynamic expressive power of prototype features. Compared to traditional methods, the CAP-Head module flexibly captures the potential correlations between the support set and query set through a multilayer combined attention mechanism, progressively optimizing the complexity of feature fusion. Moreover, CAP-Head effectively overcomes the limitations of simple element-wise multiplication fusion in the DBA-CA module, significantly enhancing the discriminative power and adaptability of prototype representations. The CAP-Head module significantly improves prototype prediction accuracy in complex few-shot learning tasks.

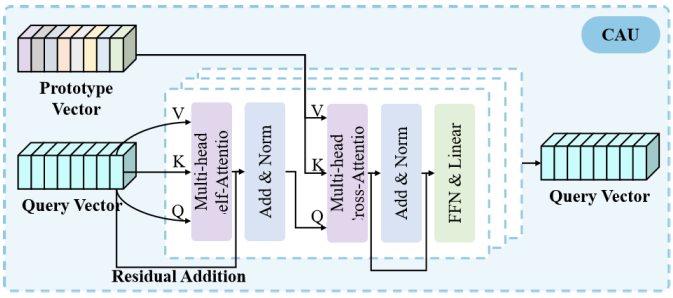


Fig. 4. Architecture of the CAU module. This module consists of a stackable composite attention computation unit, effectively capturing the complex relationships between the query set features and prototype vectors, enhancing the model's discriminative power.

Let the feature representations of the support set and query set be denoted as $Z_s \in \mathbb{R}^{N_s\times D}$ and $Z_q \in \mathbb{R}^{N_q\times D}$, where N_s and N_q represent the number of samples in the support and query sets, respectively, and D is the feature dimension. Based on the support set feature representations, we compute the initial prototype feature $p_c \in \mathbb{R}^{1\times D}$ for each class c, as shown in the following equation:

$$p_c = \frac{1}{N_c} \sum_{i=1}^{N^i_c} Z^i_s \tag{12}$$

where N_c is the number of samples belonging to class C in the support set. The set of prototypes for all classes is represented as $P = \{p_c\}^C_{c=1}$, with the tensor form $P_c = \mathrm{cat}[p_0, p_1, \ldots, p_C]$, $\dim = 0$, where C is the number of classes in the support set. The CAP-Head consists of multiple layers of combined attention units. For the lth layer of the combined attention unit, the output feature from the self-attention layer is denoted as $Z^{(l)}_q \in \mathbb{R}^{N_q\times D}$, and the

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calculation for the self-attention layer is given by the following equation:

$$Z_q^{(l)} = \text{softmax}\left(\frac{(Z_q^{(l-1)}W_Q)\cdot(Z_q^{(l-1)}W_K)^{\mathrm{T}}}{\sqrt{H}}\right)(Z_q^{(l-1)}W_V) + Z_q^{(l-1)} \tag{13}$$

where $Z_q^{(l-1)}$ is the output from the previous layer, and W_Q, W_K, and W_V are learnable linear projection matrices, with H being the scaling factor. After the self-attention operation, the updated query set features $Z_q^{(l)}$ are subjected to cross-attention computation with the prototype features P_c. In this cross-attention operation, the prototype features P_c are treated as the key and value. In contrast, the query set features $Z_q^{(l)}$ serve as the query, with the cross-attention computation described in the following equation:

$$\tilde{Z}_q^{(l)} = \text{softmax}\left(\frac{P_c \cdot P_c^{\mathrm{T}}}{\sqrt{H}}\right)Z_q^{(l)} + Z_q^{(l)}. \tag{14}$$

This process captures the global relationship between the query set and prototype features, enriching the query set features with additional intraclass information. After completing the L-layer combined attention units, the resulting feature representation $\tilde{Z}_q$ is used to calculate the predicted probability of sample q belonging to class c, as shown in the following equation:

$$\hat{y}_q^c = \frac{\exp\left(\frac{P_c\cdot\tilde{Z}_q}{\|P_c\|_2\cdot\|\tilde{Z}_q\|_2}\right)}{\sum_{c\in C}\exp\left(\frac{P_c\cdot\tilde{Z}_q}{\|P_c\|_2\cdot\|\tilde{Z}_q\|_2}\right)}. \tag{15}$$

Specifically, $(P_c \cdot \tilde{Z}_q)/(\|P_c\|_2 \cdot \|\tilde{Z}_q\|_2)$ represents the cosine similarity between the prototype feature P_c and the query feature $\tilde{Z}_q$, where $\|\cdot\|_2$ denotes the $L2$ norm of the feature vector. The model adopts the cross-entropy loss function to minimize the discrepancy between the predicted probabilities and the true labels, as shown in the following equation:

$$L_{\text{CE}} = -\frac{1}{N_q}\sum_{q=1}^{N_q}\sum_{c=1}^{N_c} y_q^c \log\left(\hat{y}_q^c\right). \tag{16}$$

By minimizing the cross-entropy loss, the model improves the similarity between the query sample and the correct class prototype while reducing confusion with other class prototypes. This process enhances the model's generalization ability in few-shot tasks.

E. Loss Function

The overall loss function of the proposed DBA-RMCL model consists of two components: the contrastive learning loss L_{CL} provided by the RSCLR module and the cross-entropy loss L_{CE} provided by the CAP-Head module. The final loss function L is a weighted combination of these two losses, balanced by preset weight parameters λ_1 and λ_2, which control the relative importance of contrastive learning and classification losses during the optimization process. The specific form of the loss function is given by (17). By adjusting these weight parameters, the model can effectively coordinate representation learning and classification tasks during training, enhancing overall performance in few-shot remote sensing image classification

$$L = \lambda_1 L_{\text{CL}} + \lambda_2 L_{\text{CE}}. \tag{17}$$

IV. Experiment and Discussion

This section describes the experimental setup and comprehensively evaluates the proposed DBA-RMCL method by comparing dataset configurations, training settings, and performance evaluation metrics. Several factors were considered: training configurations, dataset details, comparison with different approaches, and class-level results analysis. Ablation experiments further validate and optimize the key components of the technique. Visualization results effectively demonstrate the advantages and capabilities of our approach in addressing the challenges of intraclass feature diversity and interclass feature similarity in few-shot remote sensing image scene classification.

A. Dataset

We used six benchmark few-shot learning datasets to evaluate the performance of the proposed DBA-RMCL method. The data split and evaluation criteria follow the definitions established in MatchingNet [63], MKN [70], and ACL-Net [30]. Notably, we conducted few-shot learning experiments using the MEET dataset for the first time. The data partitioning and evaluation criteria were redefined in alignment with the other five datasets. The detailed splits are provided in Table I. Below are the specifics for each dataset: the NWPU-RESISC45 dataset consists of 3500 256 × 256-pixel RGB images, divided into 45 scene categories, with 700 images per category. The WHU-RS19 dataset contains 1005 256 × 256-pixel remote sensing images, categorized into 19 classes, with 50 images per class. The UC-Merced dataset, used for land-use scene recognition, includes 2100 images across 21 scene categories, with 100 images per category. The AID dataset from Google Earth includes 10 000 aerial images covering 30 scene categories, with data from various remote sensing sensors. The EuroSAT dataset, based on Sentinel satellite imagery, consists of 27 000 labeled images across 13 spectral bands and ten categories, widely used for Earth observation and geographical information improvements. The MEET dataset contains over 1.03 million unscaled remote sensing scene samples, categorized into 80 classes. We randomly sampled the classes and divided the MEET dataset into three equally sized subsets. To align with the characteristics of few-shot learning, we limited the number of samples per category to 1000 or fewer. Additionally, we created three subsets, Split1, Split2, and Split3, to conduct few-shot learning experiments. We refer to this newly constructed few-shot learning dataset as MEET-FSL. This data partition strategy enables a comprehensive evaluation of the DBA-RMCL method's few-shot learning performance across different remote sensing scenarios, supporting the validation of the model's robustness and applicability.

B. Implementation Details

We trained and evaluated the model using the PyTorch 2.4 framework on an NVIDIA GeForce RTX 4090 GPU.

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TABLE I
STANDARD SPLIT REPRESENTATION OF REMOTE SENSING SCENE CLASSIFICATION DATASETS

Dataset	D_{train}	D_{val}	D_{test}
NWPU-RESISC45	Church; Mobile home park; Stadium; Roundabout; Chaparral; Airplane; Cloud; Ship; Golf course; Meadow; Harbor; Freeway; Lake; Wetland; Baseball diamond; Island; Railway; Mountain; Sparse residential; Palace; Bridge; Desert; Sea ice; Beach; Rectangular farmland	Thermal power station; Storage tank; Terrace; Railway station; Tennis court; Snow berg; Industrial area; Runway; Overpass; Commercial area	River; Parking lot; Ground track field; Medium residential; Dense residential; Basketball court; Circular farmland; Intersection; Airport; Forest
WHU-RS19	Parking; Port; Residential; Bridge; Industrial; Mountain; Airport; Football field; Desert	Railway station; Beach; Forest; Farmland; Park	Viaduct; Pond; Meadow; River; Commercial
UC-Merced	Parking lot; Dense residential; Harbor; Chaparral; Buildings; Overpass; Agricultural; Medium residential; Baseball diamond; Freeway	Intersection; Storage tanks; Airplane; Runway; Forest	Golf course; River; Sparse residential; Tennis court; Beach; Mobile home park
AID	Bare Land; Baseball Field; Beach; Bridge; Commercial; Dense Residential; Farmland; Meadow; Pond; Port; Resort; Stadium; Airport; Desert; Railway Station	Medium residential; Mountain; Park; Parking; Playground	River; Viaduct square; Storage tanks; School; Square; Forest; Industrial; Center; Church; Sparse Residential
EuroSAT	Annual Crop; Forest; Herbaceous Vegetation; Highway; Industrial	Pasture; Permanent; Crop	Residential; River; Sea Lake
MEET-FSL	Airport; Art center; Baseball field; Bridge; Buddhist temple; Chemical plant; Church; Christmas tree farm; Cloud; Coffee plantation; Construction; Dam; Dry field; Driving school; Football field; Glacier; Highway; Lake; Landfill; Low-rise residential area; Military center; Mosque; Mountain; Office; Overpass; Paddy field; Port; Railway; Resort; Rock; Roundabout; Scrub; Solar power plant; Stadium; Substation; Theme park; Tennis court; Urban village; Vacant parking lot; Wetland.	American football field; Beach; Busy parking lot; Cemetery; Gas station; Hospital; Mangrove; Orchard; Prison; Quarry; Retail; Swimming pool; Tollbooth; Water plants; Wind power plant.	Basketball court; Brownfield; College; Commercial; Farmyard; Forest; Golf; Greenhouse; High-rise residential area; Highway service area; Hills; Meadow; Military training ground; Park; Pond; Petroleum well; Railway station; River; Sand; School; Square; Steel plant; Tea plantation; Thermal power plant; Village.

The original images were resized to 84×84 pixels before input into the backbone network. In the DBA-CA and CAP-Head modules, the scaling factor H was set to 80, and the weight ratio of the loss function's parameters λ_1 and λ_2 was set to 2:1, where λ_1 and λ_2 as defined in (17). The CAP-Head module's combination attention calculation unit consisted of five layers. Hyperparameters will be further analyzed in the ablation study. During training, we followed the training methods used in previous unsupervised contrastive learning approaches [45], applying standard data augmentation techniques such as random cropping, horizontal flipping, and color jittering. We trained scene-wise for each dataset using the SGD optimizer with a learning rate of 0.01, a momentum of 0.9, and weight decay 5×10^{-4}. Each epoch contained 500 scenes for training and 1000 scenes for validation. The query set for each class contained 20 samples, so the batch size for the 5-way 1-shot task was 105 ($5 \times (1+20)$), and for the 5-way 5-shot task, the batch size was 125 ($5 \times (5+20)$). We conducted 5-way 1-shot and 5-way 5-shot classification tasks on each dataset during testing. The final classification accuracy was obtained by averaging over 2000 randomly sampled test scenes.

C. Experimental Results and Comparisons

The advantages of our DBA-RMCL are demonstrated through a direct comparison of final classification accuracy with other state-of-the-art few-shot classification methods during the testing inference phase. The compared methods include MatchingNet [63], ProtoNet [64], Meta-SGD [65], MAML [66], RelationNet [67], DLA-MatchNet [68], SCL-MLNet [69], MKN [70], TeAW [71], MPCL-Net [72], TDNET [73], ICSFF [29], and ACL-Net [30].

Table II presents the results of the 5-way 1-shot scenario. Our DBA-RMCL model achieves the highest accuracy across all five datasets compared to other models. Precisely, on the NWPU-RESISC45, UCM, and AID datasets, the model attains accuracies of 77.15% ± 0.34%, 61.50% ± 0.21%, and 72.35% ± 0.36%, respectively, showing improvements of 1.34%, 2.95%, and 2.10% over the highest-performing models on each dataset. Table III shows the results for the 5-way 5-shot scenario, where our DBA-RMCL model achieves the highest accuracy on all datasets except for NWPU-RESISC45 and WHU-RS19. The UCM and AID datasets reach accuracies of 78.73% ± 0.25% and 90.99% ± 0.22%, respectively, outperforming the top models by 1.03% and 3.71%.

In the 5-shot setting, DBA-RMCL's performance did not meet our expectations, particularly on the NWPU-RESISC45 and WHU-RS19 datasets. The computational process of the DBA-CA module in our proposed DBA-RMCL method is influenced by the number of shots, leading to distinct behavioral patterns under different sample conditions. Specifically, when shot = 1, each class has only one support sample, meaning the model has extremely limited input features. In this scenario, the primary role of the shot branch is to model the features of a single sample, capturing intraclass feature consistency. While individual samples may lack representativeness, feature stability becomes the dominant factor.

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TABLE II

CLASSIFICATION ACCURACY OF DIFFERENT MODELS ON FIVE DATASETS IN THE 5-WAY 1-SHOT SCENARIO. BOLD RED AND UNDERLINED GREEN INDICATE THE BEST AND SECOND-BEST RESULTS, RESPECTIVELY

Method	Year	5-way 1-shot				
		NWPU-RESISC45	WHU-RS19	UC-Merced	AID	EuroSAT
MatchingNet [63]	NerusIPS-2016	61.09±0.05	56.57±0.26	54.35±0.27	49.84±0.27	61.86±0.45
ProtoNet [64]	NerusIPS-2017	63.57±0.26	67.20±0.24	51.46±0.16	50.68±0.18	76.46±0.14
Meta-SGD [65]	Arvix-2017	58.21±0.45	66.53±0.18	48.00±0.14	50.11±0.33	66.47±0.42
MAML [66]	ICML-2017	58.57±0.39	68.44±0.14	47.27±0.17	49.26±0.32	64.32±0.11
RelationNet [67]	CVPR-2018	57.34±0.33	58.82±0.13	48.15±0.28	49.13±0.28	65.63±0.27
DLA-MatchNet [68]	TGRS-2021	68.80±0.70	68.27±1.83	53.76±0.62	63.52±0.42	81.43±0.34
SCL-MLNet [69]	TGRS-2022	62.21±1.21	70.33±0.29	51.37±0.79	59.46±0.96	78.42±0.19
MKN [70]	TGRS-2022	65.84±0.89	-	58.45±0.54	57.29±0.59	-
TeAW [71]	ICASSP-2023	70.23±0.45	84.32±0.55	56.94±0.39	70.35±0.44	87.49±0.26
MPCL-Net [72]	TGRS-2023	55.94±0.04	83.74±0.25	56.46±0.21	60.61±0.43	80.93±0.44
TDNet [73]	TGRS-2023	65.85±0.53	-	-	67.48±0.51	-
ICSFF [29]	TGRS-2024	70.93±0.53	87.31±0.25	-	68.75±0.54	-
ACL-Net [30]	TGRS-2024	76.13±0.24	78.30±0.32	59.74±0.39	70.86±0.31	88.75±0.36
DBA-RMCL (ours)	-	**77.15±0.34**	**87.33±0.22**	**61.50±0.21**	**72.35±0.36**	**89.58±0.12**

TABLE III

CLASSIFICATION ACCURACY OF DIFFERENT MODELS ON FIVE DATASETS IN THE 5-WAY 5-SHOT SCENARIO. BOLD RED AND UNDERLINED GREEN INDICATE THE BEST AND SECOND-BEST RESULTS, RESPECTIVELY

Method	Year	5-way 5-shot				
		NWPU-RESISC45	WHU-RS19	UC-Merced	AID	EuroSAT
MatchingNet [63]	NerusIPS-2016	73.24±0.36	69.44±0.2	68.03±0.33	54.75±0.31	79.20±0.15
ProtoNet [64]	NerusIPS-2017	81.19±0.23	87.13±0.12	68.07±0.23	67.84±0.33	87.21±0.09
Meta-SGD [65]	Arvix-2017	76.59±0.37	79.43±0.21	63.44±0.26	60.42±0.25	81.37±0.36
MAML [66]	ICML-2017	77.32±0.42	80.42±0.13	61.92±0.18	62.97±0.11	80.62±0.19
RelationNet [67]	CVPR-2018	78.43±0.19	76.72±0.18	62.01±0.06	63.95±0.25	82.24±0.17
DLA-MatchNet [68]	TGRS-2021	81.63±0.46	79.89±0.33	63.01±0.51	83.42±0.44	89.37±0.42
SCL-MLNet [69]	TGRS-2022	80.86±0.76	86.49±0.27	68.09±0.92	76.31±0.68	88.49±0.97
MKN [70]	TGRS-2022	82.67±0.55	-	77.92±0.32	75.42±0.31	-
TeAW [71]	ICASSP-2023	85.57±0.25	91.42±0.10	77.50±0.27	84.62±0.25	92.47±0.23
MPCL-Net [72]	TGRS-2023	76.24±0.12	90.74±0.26	76.57±0.07	76.78±0.08	90.43±0.28
TDNet [73]	TGRS-2023	82.16±0.32	-	-	80.56±0.6	-
ICSFF [29]	TGRS-2024	**86.83±0.77**	**97.56±0.08**	-	87.73±0.27	-
ACL-Net [30]	TGRS-2024	86.54±0.23	90.45±0.15	74.98±0.29	85.37±0.34	95.46±0.15
DBA-RMCL (ours)	-	86.64±0.20	93.88±0.10	**78.73±0.25**	**90.99±0.22**	**95.62±0.04**

As a result, DBA-RMCL performs exceptionally well under 1-shot conditions, demonstrating its effectiveness in modeling class characteristics and extracting discriminative information even in extremely low-sample scenarios. When shot $>$ 1, the module must handle more complex feature relationships, particularly when intraclass variations are significant. With more support samples, the model gains richer class information, partially mitigating the limitations of single-sample representation. However, the increased complexity of feature relationships makes learning more challenging, requiring additional modeling of intraclass variations. This added difficulty can sometimes hinder performance gains. Consequently, while DBA-RMCL maintains strong performance in the 5-shot setting, its improvement over baseline methods is not as pronounced, merely aligning with existing SOTA methods. Overall, DBA-RMCL excels in challenging low-shot learning tasks, particularly in extreme low-sample scenarios. However, as the number of samples increases, task complexity decreases while feature relationship modeling becomes more demanding, limiting DBA-RMCL's performance gains. This suggests that while the method is highly effective for ultralow-shot tasks, its adaptability to higher-shot scenarios still has room for optimization.

To provide a more intuitive view of classification accuracy for each class, we compared the predicted classes generated by the method with the authentic courses in the five datasets. We visualized the results through confusion matrices, as shown in Fig. 5. Notably, in the 5-way 5-shot scenario, DBA-RMCL achieves nearly 100% accuracy for the 'Forest" class in the NWPU-RESISC45 dataset and nearly 100% accuracy

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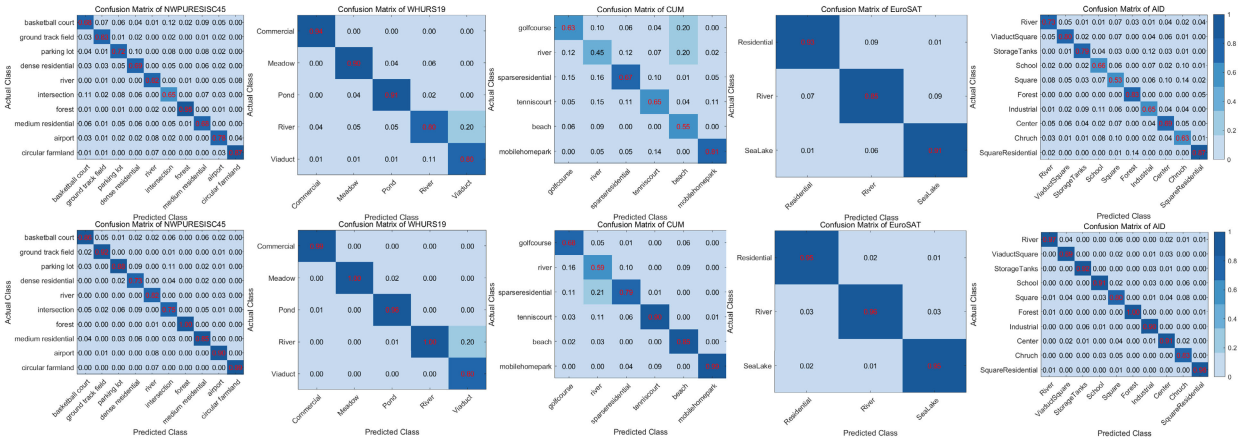


Fig. 5. Visualization of confusion matrices on the NWPU-RESISC45, WHU-RS19, UCM, AID, and EuroSAT experimental datasets. The first row shows results for a 5-way 1-shot, and the second row shows results for a 5-way 5-shot.

TABLE IV

ABLATION EXPERIMENT OF DBA-RMCL ON THE NWPU-RESISC45, WHU-RS19, AND UC-MERCED DATASETS

#	RSCLR	DBA-CA	CAP-Head	NWPU-RESISC45		WHU-RS19		UC-Merced	
				1-shot	5-shot	1-shot	5-shot	1-shot	5-shot
a				63.57±0.26	81.19±0.23	67.20±0.24	87.13±0.12	51.46±0.16	68.07±0.23
b	✓			72.44±0.16	84.11±0.13	83.24±0.15	91.35±0.37	57.83±0.19	74.89±0.27
c		✓		68.26±0.35	82.72±0.22	76.73±0.21	90.16±0.19	55.03±0.27	71.66±0.12
d			✓	65.93±0.11	82.07±0.41	71.43±0.29	88.72±0.24	53.64±0.29	70.18±0.15
e	✓	✓		75.66±0.27	85.83±0.24	85.12±0.11	93.11±0.21	59.47±0.11	76.42±0.16
f	✓		✓	73.42±0.15	84.97±0.31	84.66±0.21	92.56±0.12	58.79±0.27	75.83±0.18
g		✓	✓	70.73±0.22	83.33±0.19	78.42±0.33	90.67±0.27	56.85±0.24	73.84±0.27
h	✓	✓	✓	77.15±0.34	86.64±0.20	87.33±0.22	93.88±0.10	61.50±0.21	78.73±0.25

for the "Commercial," "Meadow," and "River" classes in the WHU-RS19 dataset. Compared to ACL-Net [30], these improvements are approximately 3%, 17%, 14%, and 3%, respectively. A detailed analysis of the class-level results will be presented in Section IV-F.

In conclusion, DBA-RMCL performs exceptionally on five publicly available remote sensing datasets in few-shot learning tasks. RSCLR encourages the model to learn high-level feature representations by maximizing the similarity of different augmented views of the same image and minimizing batch-wise distribution discrepancies. DBA-CA strengthens global features and improves intraclass compactness by capturing interclass and intraclass correlations through a dual-branch structure. CAP-Head further refines the classification performance by capturing the complex interactions between the query set and prototypes through combination attention computation units, leading to more explicit class boundaries and enhanced classification accuracy for query samples.

D. Module Ablation Experiment

We conducted comprehensive ablation studies on the NWPU-RESISC45, WHU-RS19, and UCM datasets to evaluate the contribution of each component in our DBA-RMCL framework. Table IV presents the experimental results, demonstrating each module's impact on classification performance.

1) Impact of Key Modules: Starting from the ResNet12 baseline, integrating RSCLR significantly enhanced classification accuracy. For 5-way 1-shot tasks, we observed substantial performance improvements of 13.95%, 23.87%, and 12.38% across the respective datasets. These gains validate RSCLR's effectiveness in learning high-level feature representations by optimizing interview similarity while minimizing batch distribution differences.

The DBA-CA module's dual-branch architecture further enhances global feature representation through effective management of intraclass and interclass relationships. By modeling sample-to-sample relationships, the module optimizes intraclass similarity while mitigating interclass interference, resulting in more discriminative feature representations and improved category differentiation in few-shot scenarios.

Additionally, the CAP-Head module improves performance through its composite attention calculation units, which capture complex feature interactions between query sets and prototypes. This component establishes clearer class boundaries, particularly valuable in high-dimensional feature spaces, leading to more separable prototype representations and enhanced query sample classification.

The ablation results demonstrate the effectiveness of our proposed framework, with each module contributing significantly to overall performance, particularly in few-shot learning tasks. The synergistic integration of RSCLR, DBA-CA, and CAP-Head enables DBA-RMCL to achieve superior classification accuracy and robust feature representation capabilities.

2) Comparative Analysis of Attention Mechanisms: To validate DBA-CA's effectiveness against classical attention mechanisms, we conducted comparative experiments across all

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TABLE V
COMPARISON OF CLASSIFICATION ACCURACY WITH DIFFERENT ATTENTION MECHANISMS ON THE NWPU-RESISC45, WHU-RS19, AND UC-MERCED DATASETS. BOLD RED TEXT HIGHLIGHTS THE BEST RESULTS, WHILE UNDERLINED GREEN TEXT REPRESENTS THE SECOND-BEST RESULTS

Method	NWPU-RESISC45		WHU-RS19		UC-Merced	
	1-shot	5-shot	1-shot	5-shot	1-shot	5-shot
ResNe12 (Baseline)	63.57±0.26	81.19±0.23	67.20±0.24	87.13±0.12	51.46±0.16	68.07±0.23
SE [74]	72.48±0.78	84.02±0.45	85.15±0.19	88.97±0.14	58.41±0.20	74.98±0.23
CBAM [75]	68.33±0.75	80.37±0.23	86.96±0.19	91.25±0.38	60.27±0.39	75.52±0.19
ECA [76]	66.99±0.33	80.21±0.24	82.39±0.15	86.44±0.16	60.34±0.27	75.13±0.21
SimAM [77]	65.15±0.61	81.33±0.42	78.16±0.16	86.75±0.22	56.86±0.22	72.94±0.37
GAM [39]	68.24±0.69	83.83±0.54	81.83±0.25	88.67±0.17	58.90±0.44	75.42±0.28
Without DBA-CA	73.42±0.15	84.97±0.31	84.66±0.21	92.56±0.12	58.79±0.27	75.83±0.18
DBA-CA (ours)	77.15±0.34	86.64±0.20	87.33±0.22	93.88±0.10	61.50±0.21	78.73±0.25

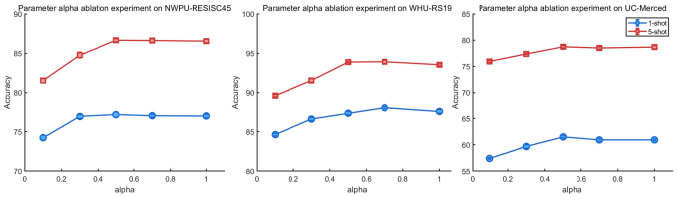


Fig. 6. Ablation study on the impact of hyperparameter loss ratio $\alpha = (\lambda_2/\lambda_1)$ on DBA-RMCL accuracy. The figure shows how varying the loss ratio between λ_2 and λ_1 affects the classification accuracy of the DBA-RMCL framework in different experimental settings.

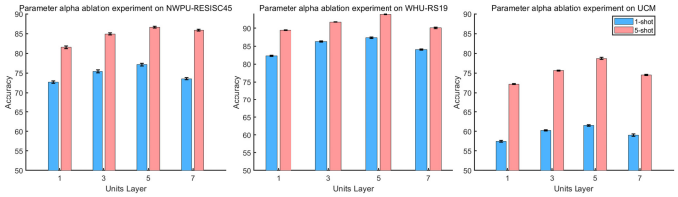


Fig. 7. Ablation study on the impact of the number of CAP-Head combination attention units on DBA-RMCL accuracy. The figure demonstrates how varying the number of combination attention units in the CAP-Head module influences the classification accuracy of the DBA-RMCL framework across different experimental conditions.

three datasets (Table V). While maintaining all other parameters constant, we isolated the impact of different attention mechanisms by replacing only the DBA-CA module. The results show that DBA-RMCL consistently achieved superior classification accuracy, while alternative attention mechanisms generally led to performance degradation.

E. Hyperparameter Sensitivity and Optimization Analysis

We conducted comprehensive experiments on the NWPU-RESISC45, WHU-RS19, and UCM datasets to optimize hyperparameters and assess their impact on model performance.

1) Loss Weight Ratio Analysis: To evaluate the balance between contrastive learning and classification objectives, we investigated the loss ratio $\alpha = (\lambda_2/\lambda_1)$, where λ_1 and λ_2 represent the weights of the contrastive loss and classification loss, respectively, as defined in (17). We tested α across $[0.1, 0.3, 0.5, 0.7, 1.0]$, with results shown in Fig. 6. Small α values, which emphasize contrastive loss, led to significant performance degradation. This occurs because excessive focus on feature similarity relationships can cause the model to overfit while neglecting classification accuracy. Optimal performance was achieved at $\alpha = 0.5$ for NWPU-RESISC45 and UCM datasets, and at $\alpha = 0.7$ for WHU-RS19. Higher α values resulted in performance plateaus or slight declines, suggesting that overemphasis on classification loss may compromise high-level feature learning and reduce model generalization. Based on these results, we selected $\alpha = 0.5$ (corresponding to a 2:1 ratio between contrastive and classification losses, that is, $\lambda_1 = 0.5$, $\lambda_2 = 1.0$) as the optimal configuration.

2) CAP-Head Layer Optimization: We next analyzed the impact of varying the number of layers (L) in the CAP-Head's composite attention calculation unit, testing L values of $[1, 3, 5, 7]$, as shown in Fig. 7. A single layer ($L = 1$) proved insufficient to capture complex prototype-query feature relationships, resulting in suboptimal performance due to limited feature interaction capacity. Optimal performance across all datasets was achieved with $L = 5$, demonstrating that multiple layers effectively enhance semantic consistency and class discriminability through progressive feature refinement. However, performance declined at $L = 7$, likely due to increased parameter complexity leading to overfitting, accumulation of noise in deeper layers, or reduced ability to capture global patterns in high-dimensional feature spaces.

Based on these experiments, we established the optimal hyperparameter configuration: loss ratio ($\alpha = 0.5$) (2:1 contrastive to classification loss ratio), and CAP-Head layers ($L = 5$). This configuration achieves an effective balance between feature representation and discriminative power while maintaining optimal model complexity across diverse datasets.

F. Visualization Result

1) CAM Analysis: We utilized class activation mapping (CAM) visualization to assess the feature attention mechanisms of DBA-RMCL across six diverse categories: Pond, Island, Palace, Storage Tanks, Airplane, and Stadium.

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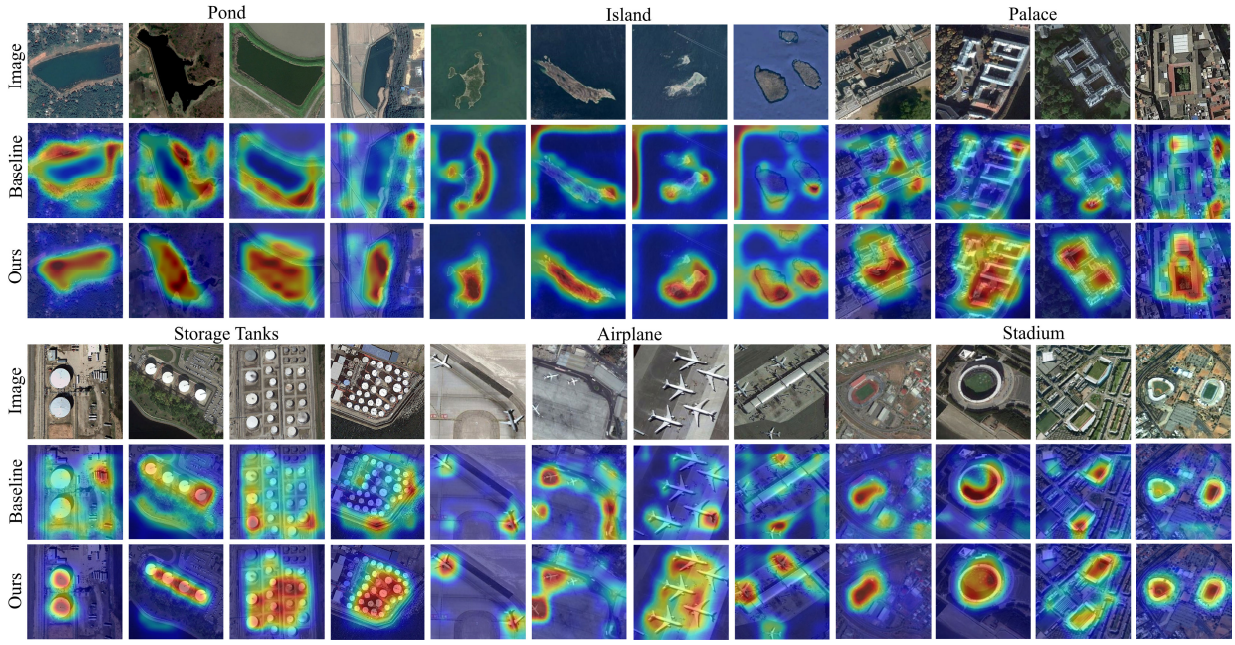


Fig. 8. CAM visualization for three (NWPU-RESISC45, WHU-RS19, and AID) classic remote sensing scenes. The red regions indicate the relevance to the target class, with darker red representing higher relevance. The blue regions indicate lower relevance. Each class is presented in three parts: the original image, the CAM features extracted by the baseline (ResNet12), and the CAM features enhanced by DBA-RMCL.

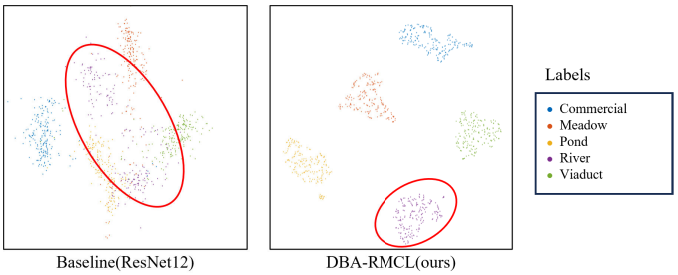


Fig. 9. t-SNE visualization results on the WHU-RS19 dataset. The figure shows the t-SNE visualization of feature distributions, demonstrating the clustering behavior of different classes using DBA-RMCL for improved feature separation.

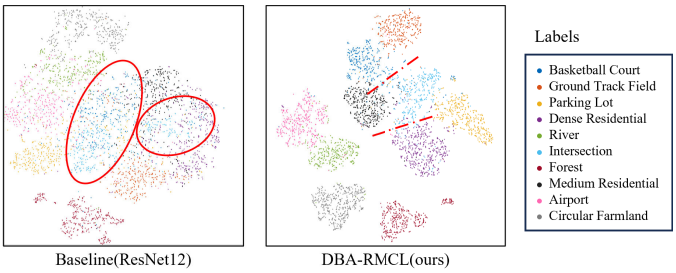


Fig. 10. t-SNE visualization results on the NWPU-RESISC45 dataset. The figure shows the t-SNE visualization of feature distributions, demonstrating the clustering behavior of different classes using DBA-RMCL for improved feature separation.

As illustrated in Fig. 8, DBA-RMCL effectively highlights regions relevant to target categories, thereby demonstrating the successful interaction between key components. Specifically, the RSCLR module enhances high-level feature representation by optimizing interview similarity and minimizing batch distribution differences. Meanwhile, the dual-branch structure of the DBA-CA module improves global feature representation by balancing intraclass compactness and interclass separability.

2) t-SNE Feature Distribution Analysis: We performed t-SNE visualization analysis on the WHU-RS19 and NWPU-RESISC45 datasets under 5-way 5-shot scenarios, as shown in Figs. 9 and 10, to investigate the feature distribution characteristics.

For the WHU-RS19 dataset, in conventional models (Fig. 9, left), the River class displays high dispersion, as highlighted by the red ellipse. Frequent misclassification occurs between the River class and similar classes such as Viaduct, Pond, and Meadow. This confusion arises from their shared color distributions and structural similarities. However, with DBA-RMCL (Fig. 9, right), the River class forms distinct clusters with clear boundaries, resulting in enhanced separation between similar classes and improved discrimination of subtle feature differences.

Regarding the NWPU-RESISC45 dataset, in conventional models (Fig. 10, left), the Intersection class shows significant scatter and overlap with classes like Medium Residential, Basketball Court, and Ground Track Field. The semantic complexity, due to the diverse scene elements, poses challenges and often causes the core intersection features to be overshadowed by peripheral elements. In contrast, with DBA-RMCL (Fig. 10, right), the Intersection class exhibits a denser and more distinct distribution, with clear boundaries between semantically similar classes. The DBA-CA module successfully addresses the challenges associated with semantic complexity.

Overall, these analyses demonstrate that DBA-RMCL effectively manages intraclass diversity and enhances interclass separation, leading to more robust feature space clustering and improved classification performance.

G. Benchmark for MEET-FSL

We conducted experiments to evaluate the performance of the DBA-RMCL method on the MEET-FSL dataset, comparing it with the ProtoNet approach under 5-way 1-shot

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TABLE VI
CLASSIFICATION ACCURACY OF DIFFERENT MODELS ON THE MEET DATASET IN THE 5-WAY 1-SHOT SCENARIO

Method	MEET-FSL		
	split1	split2	split3
ProtoNet	58.60±0.10	58.11±0.35	58.56±0.32
DBA-CA (ours)	69.64±0.22	68.88±0.21	67.98±0.25

TABLE VII
CLASSIFICATION ACCURACY OF DIFFERENT MODELS ON THE MEET DATASET IN THE 5-WAY 5-SHOT SCENARIO

Method	MEET-FSL		
	split1	split2	split3
ProtoNet	76.65±0.05	76.03±0.15	76.17±0.17
DBA-CA (ours)	86.51±0.19	85.94±0.14	85.53±0.26

and 5-way 5-shot settings. Our results establish new benchmarks for this novel few-shot learning dataset, as presented in Tables VI and VII. In the 5-way 1-shot setting, DBA-RMCL achieved accuracy rates of 69.64 ± 0.22, 68.66 ± 0.21, and 67.98 ± 0.25 on the MEET-Split1, MEET-Split2, and MEET-Split3 subsets, respectively. In contrast, ProtoNet yielded significantly lower accuracies of 58.60 ± 0.10, 58.11 ± 0.35, and 58.56 ± 0.32 on the same splits. Similarly, in the 5-way 5-shot setting, DBA-RMCL outperformed ProtoNet, achieving 86.51 ± 0.19, 85.94 ± 0.14, and 85.53 ± 0.26 on the three subsets, compared to ProtoNet's 76.65 ± 0.05, 76.03 ± 0.15, and 76.17 ± 0.17. Although DBA-RMCL has demonstrated competitive performance on the MEET-FSL dataset, further enhancing its adaptability to feature distribution shifts remains an important direction for future work. As discussed in Section IV-C, there is still room to improve its robustness, particularly in higher-shot settings or more complex feature representation learning tasks.

Further details regarding the construction and preprocessing of the MEET dataset subsets for training can be found at https://github.com/ALiAN4011/Benchmark-for-MEET-FSL

V. CONCLUSION

We propose the DBA-RMCL framework to address two key challenges in few-shot remote sensing image classification: intraclass feature variance and interclass feature similarity. Unlike existing methods, DBA-RMCL innovatively combines contrastive learning with a dual-branch architecture, focusing on enhancing the high-level representational capacity of feature learning while strengthening the discriminability between features. By precisely modeling the complex relationships between the query set and prototypes, DBA-RMCL efficiently utilizes feature representations, significantly improving classification accuracy in few-shot learning tasks. Compared to traditional methods, our approach places greater emphasis on fine-grained differentiation and efficient feature integration during the feature learning process, allowing the model to achieve stronger discriminative power across different classes. Extensive experiments on five public remote sensing datasets demonstrate the distinct advantages of our framework. Furthermore, we introduce the MEET-FSL dataset benchmark for the first time, providing a new reference for this field. Looking ahead, we plan to conduct more in-depth few-shot learning research focused on the "sense in sense" characteristics of the MEET dataset. This will involve exploring the learning and feature integration of different perceptions across branches, to improve the model's discriminative ability and robustness in more complex remote sensing image scenarios. Moreover, the modal learning of dual-branch structures still requires further in-depth research in the context of cross-domain few-shot learning.

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